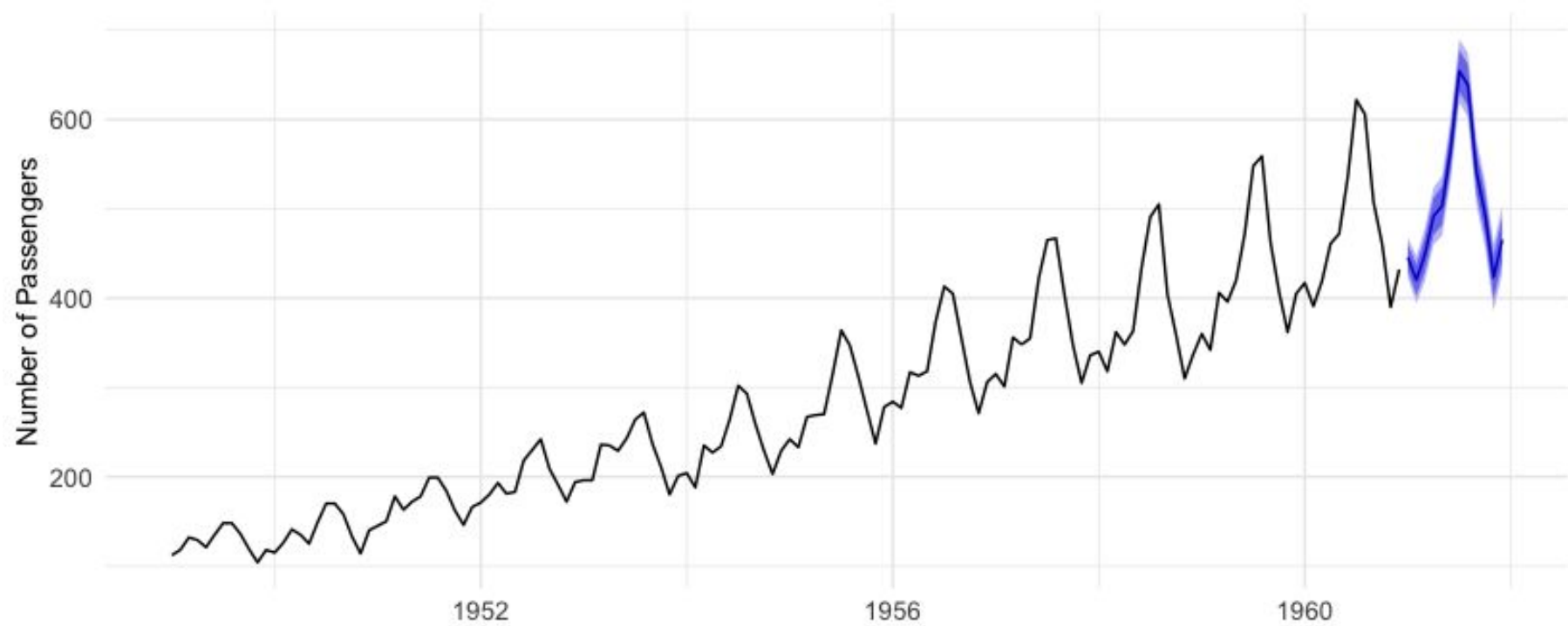




Time Series (TS)





Cleaning

- Timezone? UTC?
- Take care with daylight saving times!



Feature engineering

- It is very easy to create new columns in time series data.
- Datetime
 - Year, month, day, hour, minute, day of the week, day of the year, is weekend, is holiday, season...
 - Pandas already offers a lot of functions:
 - <https://pandas.pydata.org/pandas-docs/stable/reference/api/pandas.Timestamp.html>



Feature engineering

- Lag
 - Use data from previous times (previous hour/day...).
 - `shift` method in pandas.

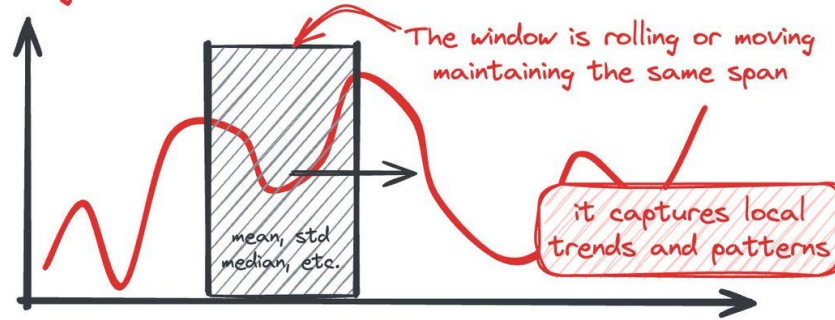


Feature engineering

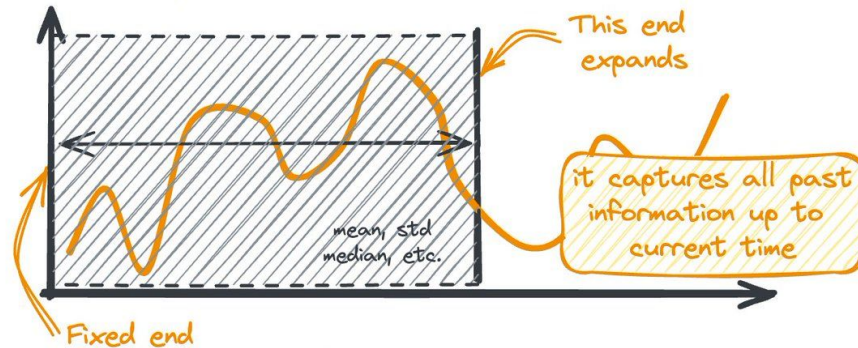
- Rolling / Expanding windows
 - Mean, weighted mean, median, mode, standard deviation of the last 10/100... elements.
 - `rolling` and `expanding` methods in pandas.

Rolling Window vs. Expanding Window

Rolling window



Expanding window

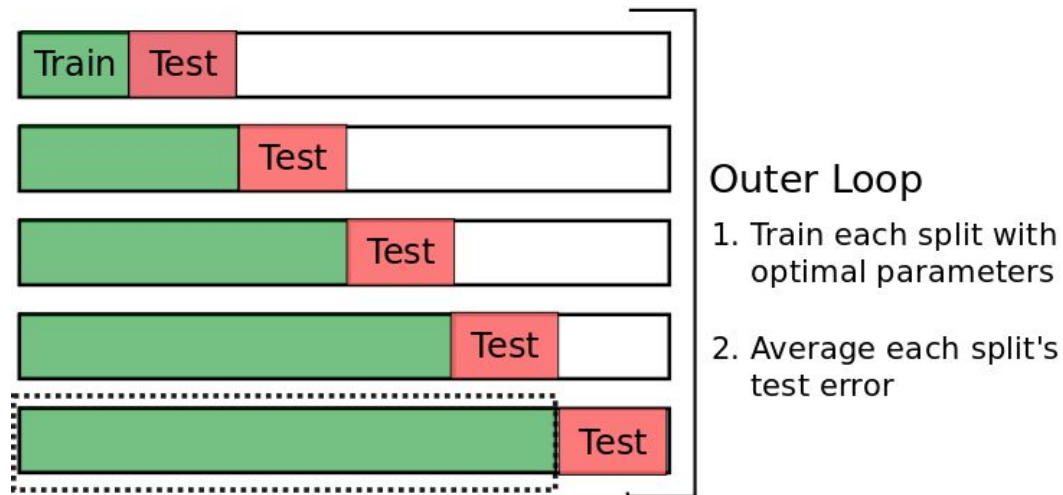




Train / Test

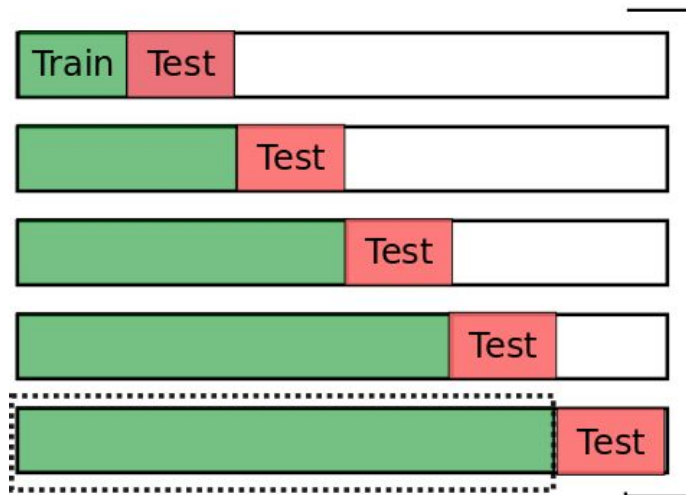


Train / Validation



Train / Validation

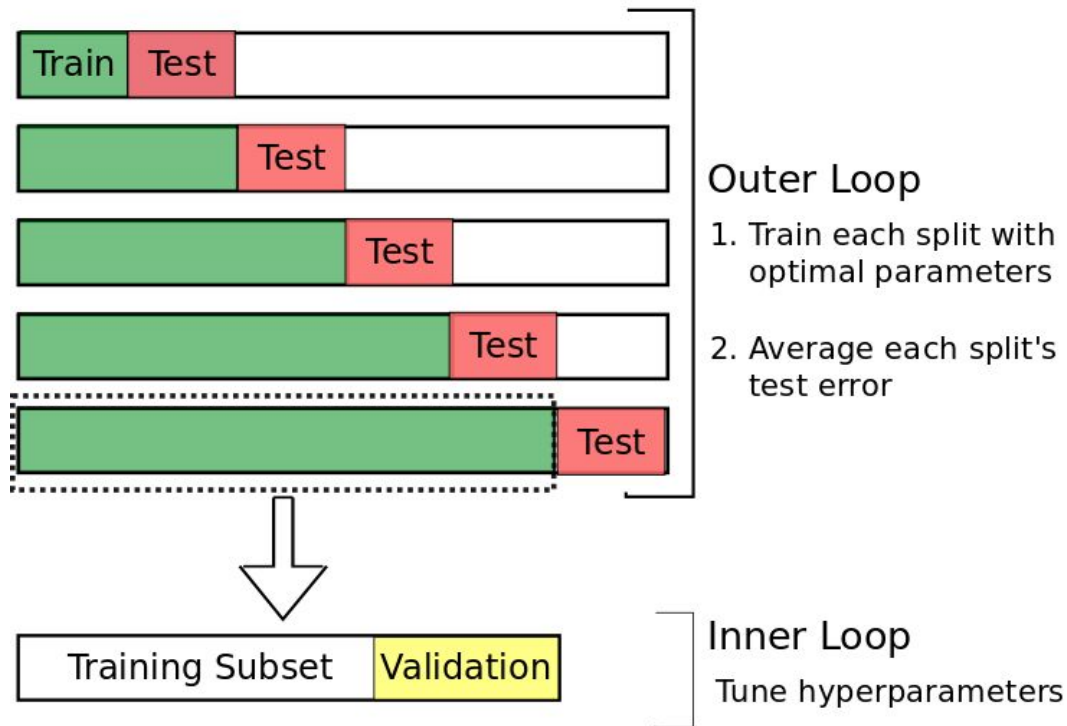
Best hyperparameters?



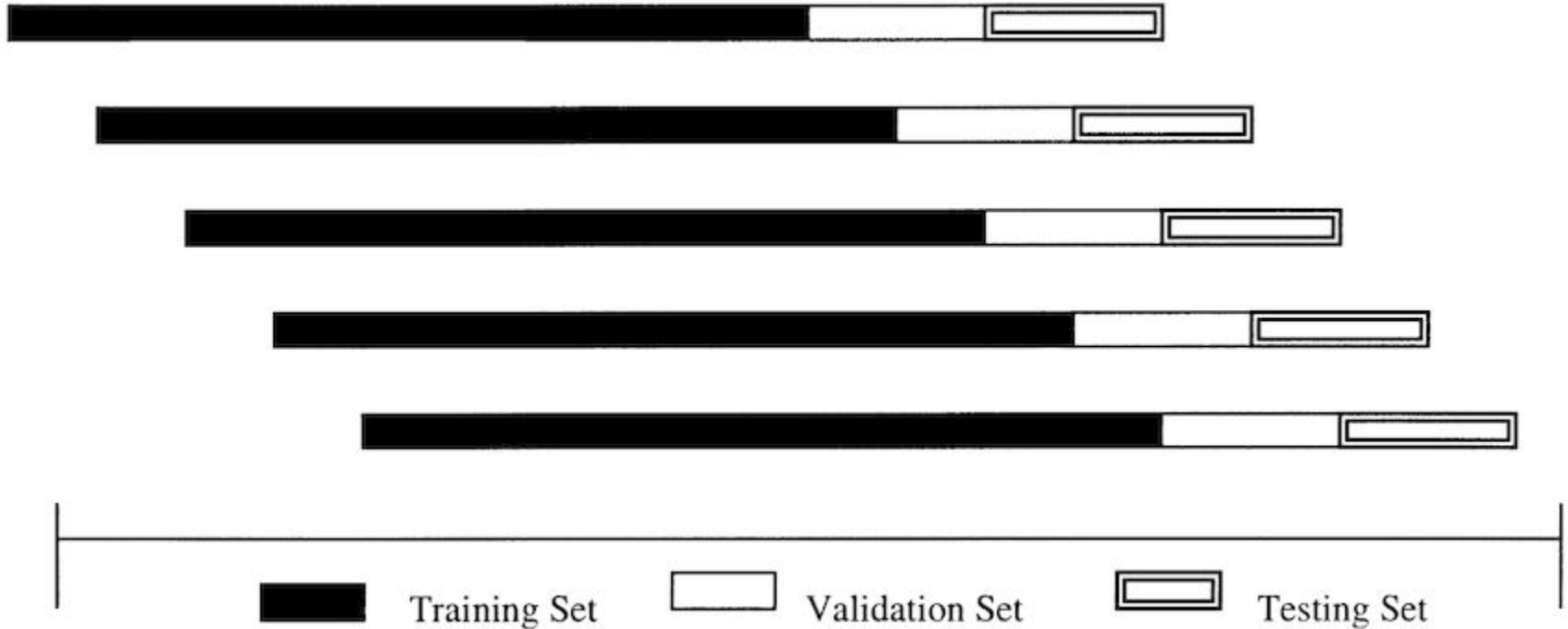
Outer Loop

1. Train each split with optimal parameters
2. Average each split's test error

Walk-forward (nested) cross-validation



Walk-forward (nested) cross-validation

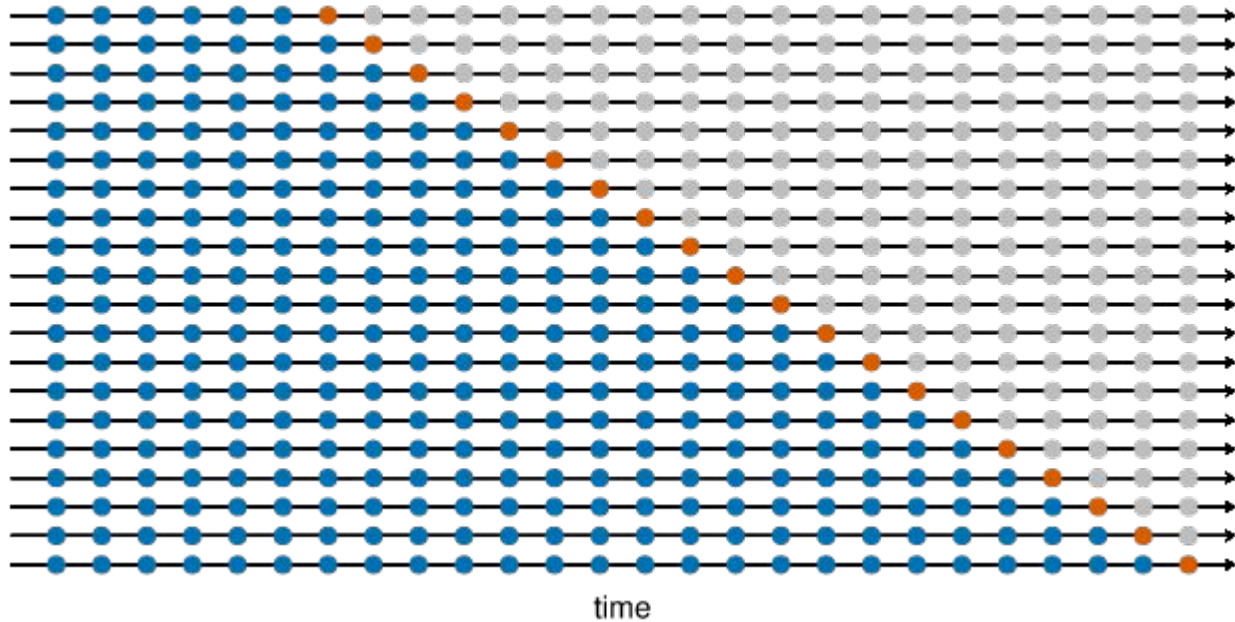




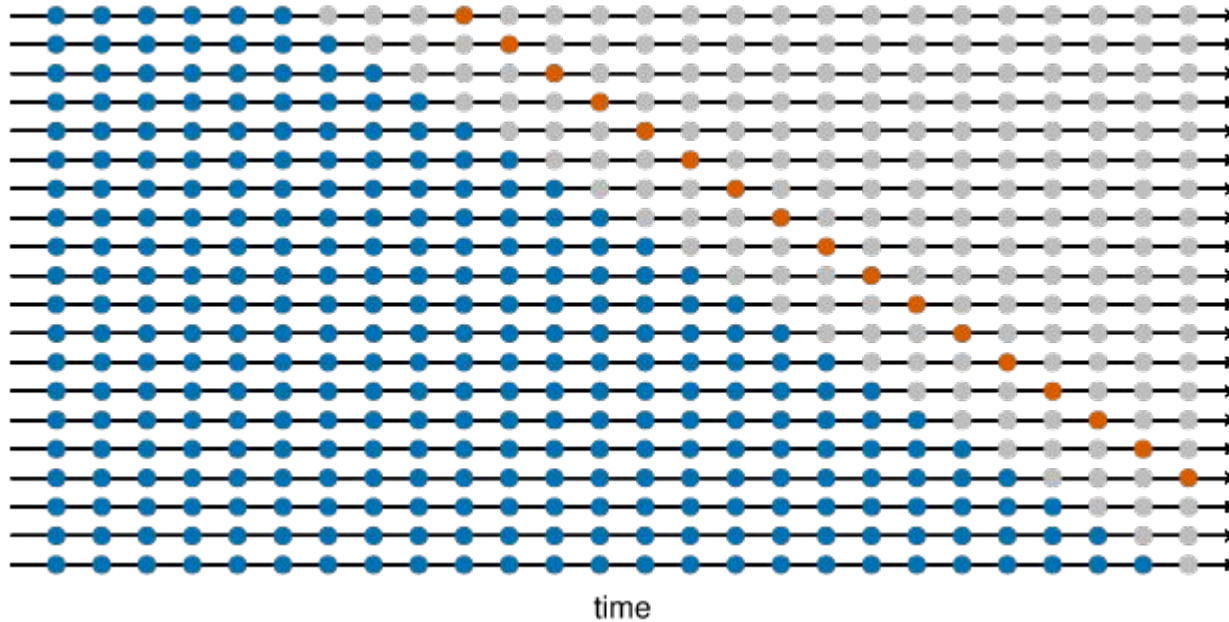
Walk-forward (nested) cross-validation

- Simulation of what would happen if we train our model every T time,
- and use the trained model for prediction in subsequent N time.

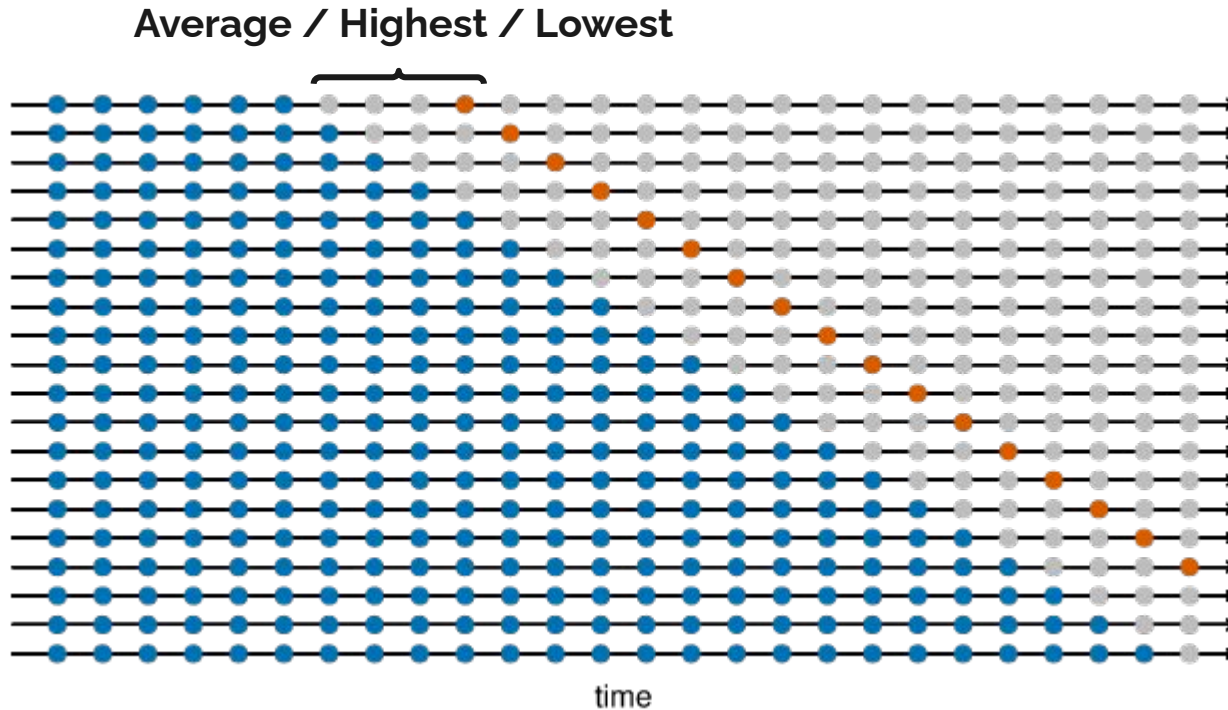
More initial training data



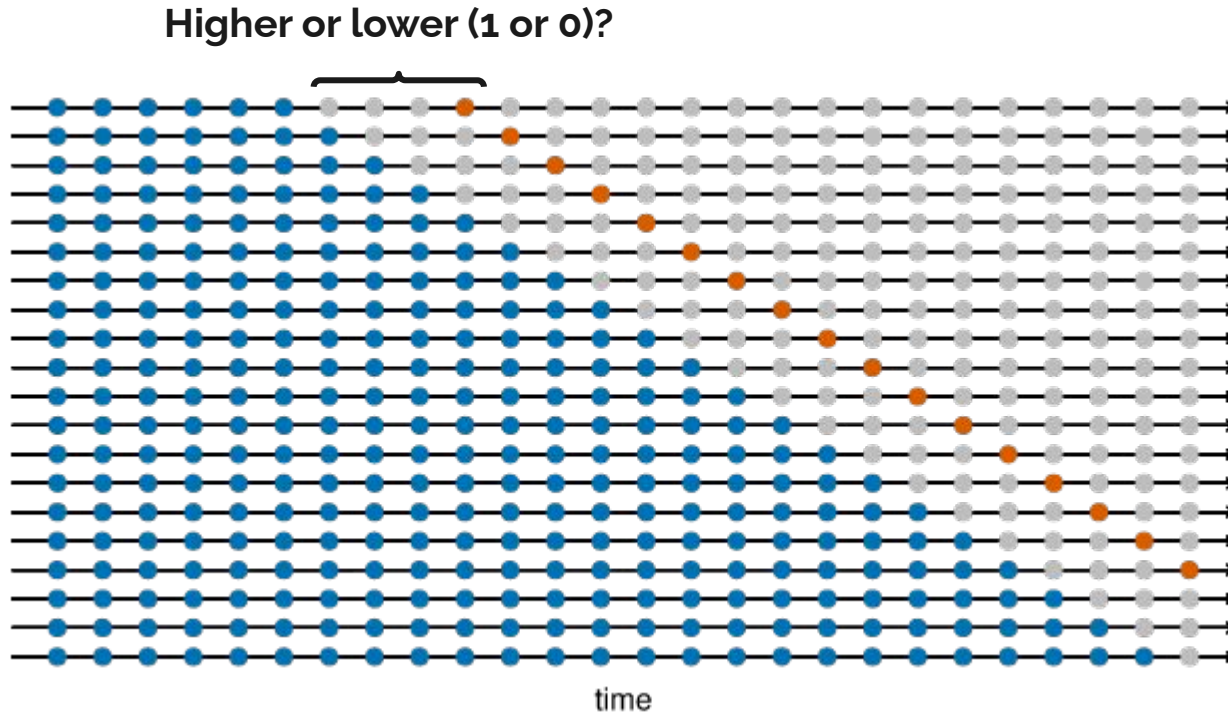
Prediction window / Forecast horizon



Prediction window / Forecast horizon



Prediction window / Forecast horizon



Univariate data



	Today's temperature
25/11/22	10
26/11/22	9
27/11/22	7
28/11/22	9
29/11/22	8

Univariate data – Create features



	-2 days temperature	-1 days temperature	Today's temperature
25/11/22	NaN	NaN	10
26/11/22	NaN	10	9
27/11/22	10	9	7
28/11/22	9	7	9
29/11/22	7	9	8

Univariate data – Target feature (1 timestep)

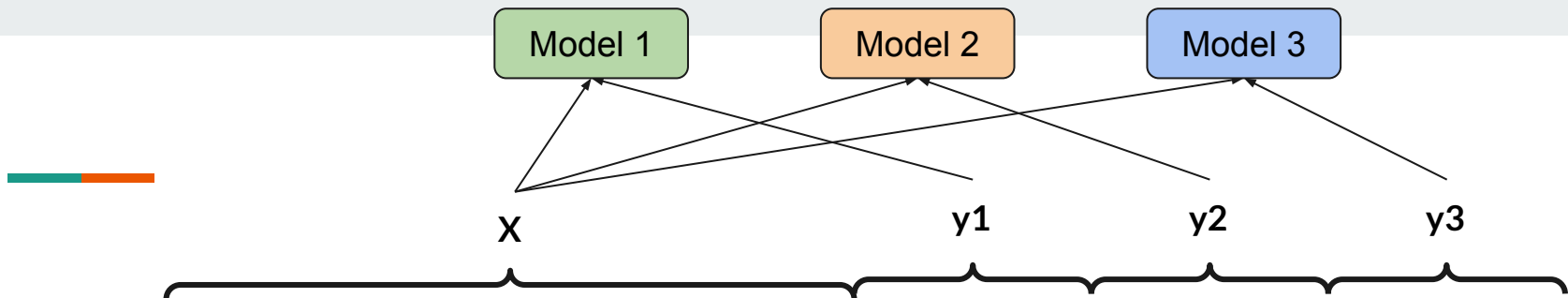


	X			y
	-2 days temperature	-1 days temperature	Today's temperature	Temperature in 1 day
25/11/22	NaN	NaN	10	9
26/11/22	NaN	10	9	7
27/11/22	10	9	7	9
28/11/22	9	7	9	8
29/11/22	7	9	8	NaN

Univariate data – Target feature (1, 2, 3 timesteps)

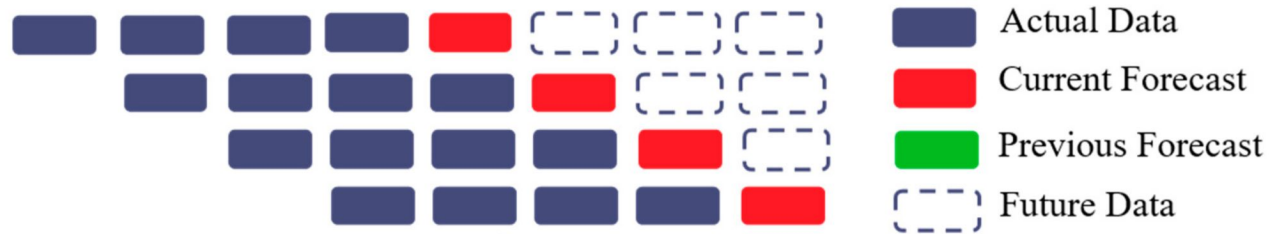


	X			y1	y2	y3
	-2 days temperature	-1 days temperature	Today's temperature	Temperature in 1 day	Temperature in 2 days	Temperature in 3 days
25/11/22	NaN	NaN	10	9	7	9
26/11/22	NaN	10	9	7	9	8
27/11/22	10	9	7	9	8	NaN
28/11/22	9	7	9	8	NaN	NaN
29/11/22	7	9	8	NaN	NaN	NaN

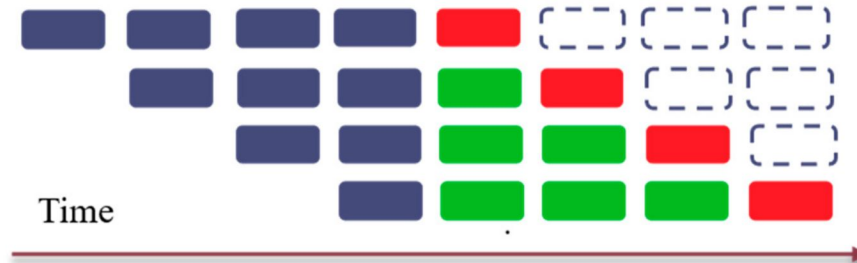


	-2 days temperature	-1 days temperature	Today's temperature	Temperature in 1 day	Temperature in 2 days	Temperature in 3 days
25/11/22	NaN	NaN	10	9	7	9
26/11/22	NaN	10	9	7	9	8
27/11/22	10	9	7	9	8	NaN
28/11/22	9	7	9	8	NaN	NaN
29/11/22	7	9	8	NaN	NaN	NaN

Prediction: multi-step ahead



a. One-Step Ahead Forecasts



Time

b. Multi-Step Ahead Forecasts



	X			Y
	-2 days temperature	-1 days temperature	Today's temperature	Temperature in 1 day
25/11/22	NaN	NaN	10	9
26/11/22	NaN	10	9	7
27/11/22	10	9	7	9
28/11/22	9	7	9	8
29/11/22	7	9	8	NaN

	X			y	
	-2 days temperature	-1 days temperature	Today's temperature	Temperature in 1 day	
25/11/22	NaN	NaN	10	9	} Train
26/11/22	NaN	10	9	7	
27/11/22	10	9	7	9	
28/11/22	9	7	9	8	
29/11/22	7	9	8	Predict 1	

	X			y	
	-2 days temperature	-1 days temperature	Today's temperature	Temperature in 1 day	
25/11/22	NaN	NaN	10	9	} Train
26/11/22	NaN	10	9	7	
27/11/22	10	9	7	9	
28/11/22	9	7	9	8	
29/11/22	7	9	8	Predict 1	
30/11/22	9	8	Predict 1	Predict 2	

	X			y	
	-2 days temperature	-1 days temperature	Today's temperature	Temperature in 1 day	
25/11/22	NaN	NaN	10	9	} Train
26/11/22	NaN	10	9	7	
27/11/22	10	9	7	9	
28/11/22	9	7	9	8	
29/11/22	7	9	8	Predict 1	
30/11/22	9	8	Predict 1	Predict 2	
31/11/22	8	Predict 1	Predict 2	Predict 3	



Multivariate

- Similar to univariate, applying the same principles to all features.